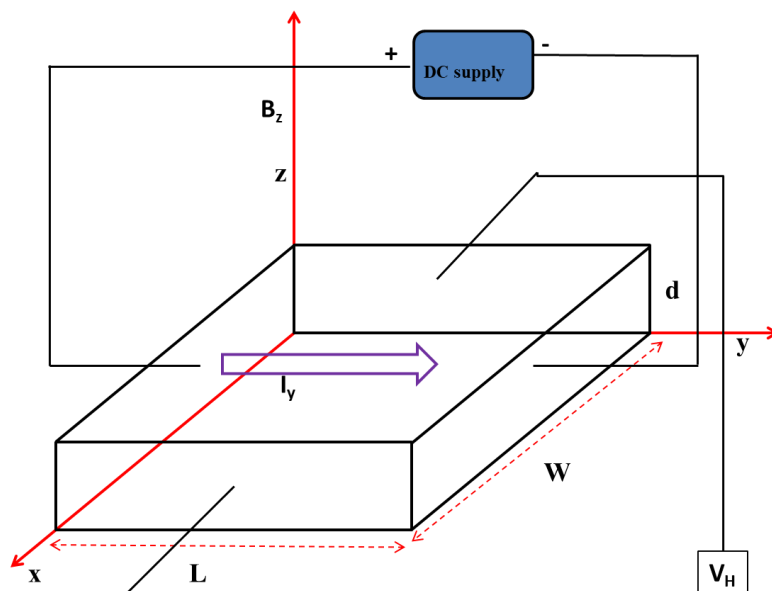


Hall Effect:

The Hall Effect is a phenomenon where a solid (metal or Semiconductor) material, conducting an electric current, generates a transverse electric field when exposed to a magnetic field perpendicular to the current.

Suppose an electric an electric field E is applied across the two faces of the semiconductor slab (DC Supply). The slab length is L width W and thickness d . Semiconductor slab is placed in an external magnetic field B_z , which is along z direction. When the slab is connected through the DC power supply a current I , is start flowing across the semiconductor slab as shown in the Figure.



Since current flows due to motion of the electrons/holes and these electrons will experience magnetic force $F = e v_y B_z \dots \dots \dots (1)$

Due to this magnetic force, electrons starts accumulating at one surface and an induced electric field starts developed resulting a Hall voltage V_H .

When the electric force due to developed Hall electric field $E_H = \frac{V_H}{W} \dots \dots \dots (2)$ becomes equal to magnetic force given in equation (1).

Then $e E_y = e E_H = e v_y B_z$

$$E_H = v_y B_z \dots \dots \dots (2)$$

$$V_H = W v_y B_z \dots \dots \dots (3)$$

Since $v_y = v_d$ (drift velocity)

For n-type semiconductor

Current density $I_y = e n_e v_d A = e n_e v_d W d$

Where $A = W d$ refer to above figure

Using above equations we get

$$\text{Hall voltage } V_H = \frac{I_y B_z}{e n_e d} \dots \dots \dots (4)$$

Hall Effect sensors: It is a kind of magnetic sensor which can be used for detecting the strength and direction of magnetic field produced from the permanent magnet or an electromagnet.

Magnetic Switches: Magnetic switch consists of flat contact plates sealed in a glass tube filled with an inert gas. When a permanent magnet approaches, the contact plates begin to attract each other and close. When the permanent magnet is removed, the contact plates instantly demagnetize and return to their original position.

Magnetic switches excel in challenging environments, effortlessly handling dirt, humidity, gas, dust, chips, and more. With a stable switching point and a remarkable reproducibility error of just 0.01 mm, they guarantee reliable performance.

Absorption: When an atom situated in a lower energy level absorbs a photon with a frequency of $h\nu$, its transitions to a higher energy level is called absorption.

Spontaneous emission: An atom positioned in a higher energy level can spontaneously transition to a lower level, emitting a photon with a frequency of $h\nu$ if the transition between E_2 and E_1 is radiative. This emitted photon has a random direction and phase. Such transition is called spontaneous emission.

Stimulated emission: An incoming photon prompts an atom in an upper energy level to decay, releasing a "stimulated" photon that shares identical properties with the incident photon. The term "stimulated" emphasizes that this radiation occurs only in the presence of an incident photon. The amplification results from the similarities between the incident and emitted photons.

